A MICRO PROJECT REPORT

ON

"AI Student Career Recommendation System using CSP + Forward/Backward Chaining"

Subject: Foundation of Artificial Intelligence

(Course Code: 2ADPC204)

Submitted in partial fulfilment of the requirement for

the award of the Degree in SY (AI & DS).

SUBMITTED BY

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CERTIFICATE

This is to certify that Ms. Sakshi Ajay Patil of SY B. Tech (AI & DS), Roll No. 2051, of Annasaheb Dange College of Engineering and Technology, Ashta, has successfully completed the micro project entitled “AI Student Career Recommendation System using CSP + Forward/Backward Chaining” in the subject Foundation of Artificial Intelligence (Course Code: 2ADPC204) for the academic year 2026–2027 as prescribed in the curriculum.

Place: _____

Date: _____

Subject Teacher

(Signature)

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MICRO PROJECT REPORT

AI STUDENT CAREER RECOMMENDATION SYSTEM USING CSP + FORWARD/BACKWARD CHAINING

1. Aims / Benefits of the Micro Project

Aim: To develop an AI-based career recommendation system that suggests suitable career paths for students by representing their interests, skills, subjects, and preferences as a Constraint Satisfaction Problem (CSP), then applying forward and backward chaining to infer recommendations.

• Recommends career options based on student profile and eligibility constraints.

• Demonstrates CSP variables, domains, and constraints.

• Uses forward chaining to derive careers from known student facts.

• Uses backward chaining to verify whether a target career is supported by the facts.

• Improves logical reasoning, explainability, and decision support.

2. Course Outcomes Achieved

Understood basic concepts of Artificial Intelligence and knowledge-based systems.

Identified variables, domains, and constraints for student career selection.

Applied forward chaining and backward chaining to infer conclusions.

Used rules to connect interests and skills with suitable career paths.

Improved logical thinking, testing, and debugging skills.

3. Literature Review

Career recommendation systems help students explore pathways using information such as interests, academic subjects, skills, and goals. A CSP models a problem using variables, domains, and constraints. Rule-based reasoning complements CSP: forward chaining starts from known facts and fires applicable rules, while backward chaining starts from a proposed goal and checks whether supporting facts and rules establish it. Combining these methods can provide recommendations with understandable reasons.

4. Actual Methodology Followed

Collected a sample student profile containing interests, strong subjects, skills, and preferred work style.

Defined career options such as AI/ML Engineer, Data Analyst, Software Developer, and Cybersecurity Analyst.

Created CSP variables for interests, skills, subject strengths, and career choice.

Defined domains for each variable and constraints connecting profile attributes to career requirements.

Stored student attributes as facts in a small knowledge base.

Applied forward chaining to infer possible career paths from the known facts.

Applied backward chaining to test a selected career goal against rules and facts.

Displayed the recommended career and the rule-based reasons behind it.

5. Actual Resources Used

Hardware: Computer or laptop, keyboard and display.

Software: Python 3; Visual Studio Code or another Python editor; terminal/command prompt.

Other resources: Foundation of AI class notes, Python documentation, sample student profiles, and testing inputs.

Problem Definition

Input: Student interests, subject strengths, skills, and preferred work style. Output: One or more career recommendations that satisfy the profile constraints.

CSP Components

Variables: Interest, StrongSubject, Skills, WorkStyle, Career.

Domains: Values such as AI, mathematics, programming, analytics, communication; career options from a defined list.

Constraints: Recommended career should match relevant interests and skills; required subjects/skills should be present; conflicting preferences should be avoided where possible.

6. Algorithm: CSP and Forward/Backward Chaining

Algorithm

Start and read the student profile.

Create CSP variables, domains, and constraints.

Remove career options that violate hard constraints.

Store profile attributes as facts in the knowledge base.

Forward chaining: repeatedly apply rules whose conditions match current facts and add their conclusions.

Backward chaining: choose a target career and recursively check whether its supporting rule conditions can be proved.

Combine the CSP-filtered options with rule-based conclusions.

Display career recommendations and reasons; stop.

Pseudocode

```
INPUT student_profile
facts = attributes_from(student_profile)
candidates = all_careers

FOR each career IN candidates:
    IF career violates a hard constraint:
        remove career

REPEAT:
    changed = false
    FOR each rule IF rule conditions are in facts:
        add rule conclusion to facts
        changed = true
UNTIL changed is false

FOR each candidate career:
    use backward_chaining(career_goal, facts, rules)
    if goal is proven and candidate satisfies CSP:
        recommend candidate

DISPLAY recommendations and supporting reasons
```

Forward Chaining

Forward chaining is data-driven. It begins with known facts, checks rules, and adds new conclusions. Example: interest_in_AI + strong_in_math + programming_skill → possible_AI_ML_career.

Backward Chaining

Backward chaining is goal-driven. It begins with a career goal and checks whether the required conditions can be proved from facts and rules.

7. Program / Implementation

The following Python implementation uses profile facts, CSP-style constraints, and simple forward/backward chaining rules. It is a console-based prototype.

```
# AI Student Career Recommendation System
# CSP + Forward and Backward Chaining

RULES = [
    ({"interest_ai", "strong_math", "programming"}, "AI/ML Engineer"),
    ({"interest_data", "strong_math", "analytical"}, "Data Analyst"),
    ({"programming", "problem_solving", "interest_software"}, "Software Developer"),
    ({"networking", "security_interest", "problem_solving"},
     "Cybersecurity Analyst"),
]

def forward_chaining(facts):
    inferred = set(facts)
    recommendations = set()
    changed = True
    while changed:
        changed = False
        for conditions, career in RULES:
            if conditions.issubset(inferred) and career not in recommendations:
                recommendations.add(career)
                inferred.add("career:" + career)
                changed = True
    return recommendations

def backward_chaining(goal, facts):
    for conditions, career in RULES:
        if career == goal:
            return conditions.issubset(set(facts))
    return False

def csp_filter(careers, preferred):
    # Hard constraint: keep only careers supported by profile rules.
    return [career for career in careers if career in preferred]

def main():
    print("AI Student Career Recommendation System")
    print("Enter facts separated by commas.")
    print("Examples: interest_ai, strong_math, programming")
    facts = {x.strip().lower() for x in input("Student facts: ").split(",")}
    inferred = forward_chaining(facts)
    valid = csp_filter(sorted(inferred), inferred)

    if not valid:
        print("No exact match. Build more skills or add profile facts.")
    else:
        print("\nRecommended careers:")
        for career in valid:
            print("-", career)
            print("  Goal verified:", backward_chaining(career, facts))

if __name__ == "__main__":
    main()
```

Main Features

• Uses student facts as input.

• Applies rules through forward chaining.

• Checks a career goal using backward chaining.

• Uses CSP-style filtering to keep supported career options.

• Shows recommendations and whether each goal is verified.

8. Outputs of the Micro Project — Sample Output

Example input:

```
interest_ai, strong_math, programming
```

Expected output:

```
AI Student Career Recommendation System
Enter facts separated by commas.
Examples: interest_ai, strong_math, programming
Student facts: interest_ai, strong_math, programming

Recommended careers:
- AI/ML Engineer
  Goal verified: True
```

Explanation: The three input facts satisfy the rule for AI/ML Engineer. Forward chaining derives this career, and backward chaining verifies that the conditions for the selected career are present.

Another example: facts `interest_data, strong_math, analytical` lead to the recommendation `Data Analyst`.

Website / Console Interface and Testing

This version is a Python console prototype rather than a website interface. It accepts profile facts, runs the rule engine, and prints career options with verification results.

Test Cases

Test	Input facts	Expected recommendation
1	interest_ai, strong_math, programming	AI/ML Engineer
2	interest_data, strong_math, analytical	Data Analyst
3	programming, problem_solving, interest_software	Software Developer
4	networking, security_interest, problem_solving	Cybersecurity Analyst
5	interest_music	No exact match message

Note: These are sample test cases for the included prototype. Recommendations depend on the facts supplied; they are guidance for exploration, not a final or guaranteed career decision.

9. Skills Developed

• Python programming and rule-based reasoning.

• Understanding CSP variables, domains, and constraints.

• Forward chaining and backward chaining.

• Knowledge representation using facts and IF–THEN rules.

• Testing, debugging, and explaining system output.

• Designing an explainable AI decision-support prototype.

10. Applications of the Micro Project

Student career guidance and exploration.

Course and skill pathway suggestions.

Academic advising support.

Demonstrations of CSP and inference methods.

AI coursework, practical demonstrations, and knowledge-based systems.

Conclusion

The AI Student Career Recommendation System demonstrates how career selection can be represented as a Constraint Satisfaction Problem and supported by rule-based inference. CSP constraints help filter options according to a student profile. Forward chaining derives possible careers from known facts, while backward chaining checks whether the conditions for a target career are satisfied. The prototype provides simple, explainable recommendations that can help students explore options; final decisions should also consider personal goals, further study, and professional guidance.

References

Python Documentation: https://docs.python.org/3/

Russell, S. and Norvig, P., Artificial Intelligence: A Modern Approach, Pearson.

Foundation of Artificial Intelligence class notes on CSP and inference.

Course notes on forward chaining and backward chaining.